

Project Proposal: Spotify Recommendation System
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Introduction:

I have been a dedicated Spotify user for about a decade, and I've witnessed the evolution of their music recommendation algorithms. From the dawn of the platform to the introduction of cutting-edge features like the AI DJ and daylists, Spotify has continually strived to enhance the user experience by delivering personalized music recommendations. The AI DJ, for instance, has been a game-changer, dynamically curating playlists that seamlessly blend new discoveries with cherished favorites, all categorized by genre and refreshed every few songs.

Despite the innovative strides Spotify has taken, the recommendations generated by these algorithms didn't always align with my musical preferences. This realization prompted me to delve deeper into the intricacies of my listening habits, leading to the conceptualization of a personal Spotify recommendation algorithm.

The inspiration for this project stemmed from my recent exploration of Spotify Wrapped, an annual feature that encapsulates a user's most-played songs, genres, and artists over the past year. Analyzing this data, I contemplated the feasibility of creating a recommendation system tailored to my tastes. The goal is to leverage cluster analysis methods, an unsupervised machine learning approach, to group my liked songs based on their metadata.

The algorithm would autonomously categorize my music library into clusters based on shared characteristics, such as genre, tempo, or mood. This approach differs from Spotify's existing algorithms, which rely on broader patterns and user behavior across the platform. By focusing on my listening history, the algorithm aims to provide genre-specific recommendations that resonate with the nuanced preferences inherent in my music taste.

The use of cluster analysis introduces an intriguing dimension to the project. I am particularly curious to uncover the averages and patterns within the clusters generated by the model. This exploration into the unsupervised learning space adds an element of discovery, as it unveils hidden relationships and connections among my favorite songs that may not be immediately apparent through traditional playlist curation methods.

APPROACH:

The key to this project is understanding and enhancing my music recommendations on Spotify. I gathered the metadata of songs I've liked over the past three years. This required a visit to Exportify, a scraping website that, upon authorization, provided a comprehensive dataset of my liked songs. The dataset encompasses various numerical descriptors such as danceability,

energy, key, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and time signature.

The significance of these variables lies in their ability to represent the musical attributes of each liked song quantitatively. These variables allow for analysis of my music preferences, extending beyond traditional genre categorizations. The objective is to employ cluster analysis methods to group these songs based on their metadata, unraveling patterns and associations that might not be immediately apparent through manual inspection.

Simultaneously, I intend to gather similar data from a random Spotify playlist comprising more than 1000 songs. The goal is to apply the same clustering methodology to categorize these songs based on their attributes. The ensuing step involves comparing the clusters generated from my liked songs with those from the random playlist, providing insights into the distinctiveness of my music preferences.

A critical phase of the project involves testing the efficacy of the recommendation algorithm. To achieve this, I select a subset of the top 100 recommended songs and listen to each, subsequently rating them based on personal enjoyment. This evaluation process serves as a litmus test for the algorithm's ability to predict songs that align with my preferences accurately.

In parallel, I'm incorporating tree classifiers into the evaluation framework. This necessitates some data engineering to create a hybrid dataset. By merging random songs with my liked playlist, I aim to establish a binary classification system – liked or not liked. This binary target variable forms the basis for training tree classifiers, adding a machine learning dimension to the project.

The tree classifiers provide an alternate perspective on the recommendation process, allowing for a comparison of the unsupervised clustering approach with a supervised learning methodology. The model's effectiveness will be gauged by its ability to accurately classify songs into liked and not liked categories, offering a quantitative measure of its predictive capabilities.

I plan to listen to and rate the top hundred songs recommended by the tree classifiers to enrich the evaluation process further. This comprehensive assessment, encompassing both clustering and classification methodologies, aims to provide a holistic understanding of the strengths and potential areas of improvement for the recommendation algorithm.

Data Summary

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summary(df)

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Spotify ID Length:977 Class :character Mode :character	Artist IDs Length:977 Class :character Mode :character	Track Name Length:977 Class :character Mode :character	Album Name Length:977 Class :character Mode :character	Artist Name(s) Length:977 Class :character Mode :character	Release Date Length:977 Class :character Mode :character	Duration (ms) Length:977 Class :character Mode :character
Popularity Length:977 Class :character Mode :character	Added By Length:977 Class :character Mode :character	Added At Length:977 Class :character Mode :character	Genres Length:977 Class :character Mode :character	Danceability Length:977 Class :character Mode :character	Energy Length:977 Class :character Mode :character	Key Min. : 0.000 1st Qu.: 2.000 Median : 5.000 Mean : 5.172 3rd Qu.: 8.000 Max. : 11.000
Loudness Min. : -20.001 1st Qu.: -7.246 Median : -5.597 Mean : -5.901 3rd Qu.: -4.393 Max. : 0.943	Mode Min. : 0.0000 1st Qu.: 0.0000 Median : 1.0000 Mean : 0.6407 3rd Qu.: 1.0000 Max. : 10.0000	Speechiness Min. : -5.97000 1st Qu.: 0.03720 Median : 0.05100 Mean : 0.08593 3rd Qu.: 0.08940 Max. : 0.59200	Acousticness Min. : 0.0000 1st Qu.: 0.0162 Median : 0.0693 Mean : 0.1900 3rd Qu.: 0.2770 Max. : 0.9710	Instrumentalness Min. : 0.000000 1st Qu.: 0.000000 Median : 0.000000 Mean : 0.030994 3rd Qu.: 0.000095 Max. : 0.947000	Liveness Min. : 0.0304 1st Qu.: 0.1040 Median : 0.1310 Mean : 0.1807 3rd Qu.: 0.2220 Max. : 0.7780	Valence Min. : 0.0000 1st Qu.: 0.2220 Median : 0.3500 Mean : 0.3783 3rd Qu.: 0.5120 Max. : 0.9690
Tempo Min. : 0.268 1st Qu.: 104.443 Median : 127.923 Mean : 123.817 3rd Qu.: 140.063 Max. : 203.759	Time Signature Length:977 Class :character Mode :character					

Model Application:

Decision tree models stand out as one of the most powerful and versatile tools in the realm of machine learning, especially when it comes to classification tasks. These models are adept at handling a wide range of output or target variables, making them a popular choice in various domains. In my current project, which revolves around refining music recommendations on Spotify, classifying tree models emerge as a key player in understanding and enhancing the recommendation algorithm.

The fundamental principle behind classifying tree models is their ability to make decisions based on input features, effectively creating a tree-like structure of decisions and outcomes. In the case of music classification, these decision trees can be employed to make crucial business decisions related to user preferences. By analyzing the features of songs, such as danceability, energy, key, and tempo, the model can intelligently classify music, aiding in the recommendation process. For instance, a decision tree could discern whether a user will likely enjoy a song based on its attributes, ultimately contributing to the refinement of personalized playlists.

In my specific project, I leverage classifying tree models to evaluate how well my algorithm classifies music according to my preferences. The model analyzes the metadata of songs I have liked over the past three years, creating a decision tree based on these features. The goal is to assess the model's accuracy in predicting which songs align with my taste, thus refining the recommendation algorithm.

On the other hand, clustering represents a different paradigm within the realm of machine learning. Unlike classifying trees, clustering is an unsupervised learning method. It is particularly useful when the goal is to uncover hidden patterns or groupings within data without predefined categories. In my project, I employ clustering to group the songs I like based on their metadata.

This unsupervised approach allows the model to identify structures within the data, revealing patterns that were not apparent through manual inspection.

However, clustering has challenges, especially when dealing with multiple dimensions. In music metadata, numerous features, such as danceability, energy, and tempo, create a multidimensional space. Visualizing and interpreting this space can be intricate, prompting the need for dimension-reduction techniques. Dimension reduction simplifies data representation by transforming it into a lower-dimensional space while preserving its essential characteristics.

As I delve into the clustering aspect of the project, I anticipate the need to employ dimension-reduction techniques to visualize and interpret the clusters effectively. Techniques like Principal Component Analysis (PCA) or t-distributed Stochastic Neighbor Embedding (t-SNE) could prove invaluable in simplifying the complexity of the multidimensional space and aiding in the interpretation of clustering results.